Introduction to Microeconomics Fall 2025

SEMINAR #3

19 NOVEMBER 2025

FRANCESCO CASALENA FRANCESCO.CASALENA@UNIGE.CH

Exercise 1

Consider a perfectly competitive market. The supply and demand functions are given by $Q^D = 100 - 2P$ and $Q^S = -20 + 8P$, respectively.

1. Determine the equilibrium price and quantity, producer surplus and the consumer surplus.

A tax is introduced. It amounts to 10.

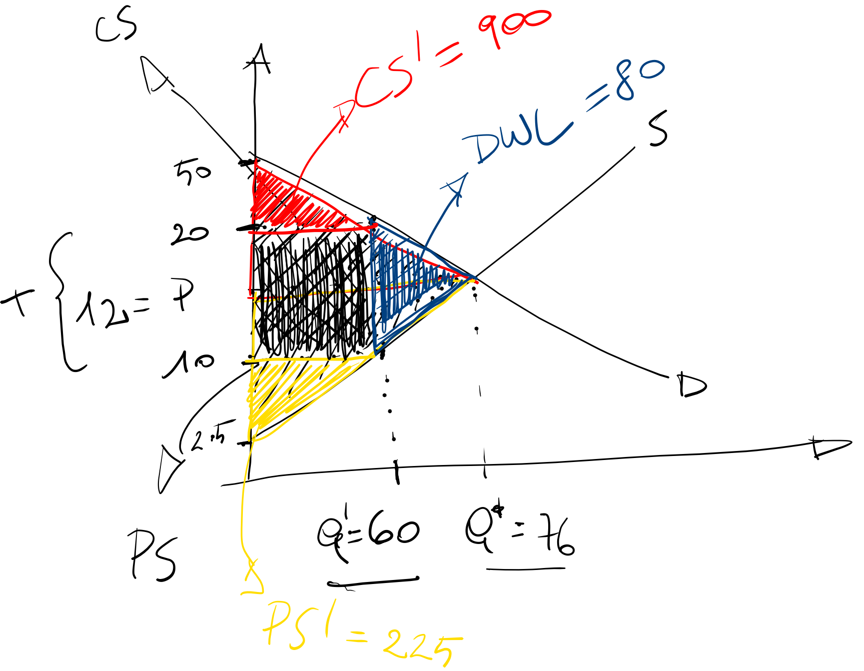
2. What is the quantity traded?

3. What is the price received by producers and that paid by consumers?

4. What is the new consumer surplus? And that of the producer? Calculate the deadweight loss.

$$Q^D = 100 - 2P$$
$$Q^S = -20 + 8P$$

$$100 - 2P = -20 + 8P$$
$$P^* = 12$$
$$Q^* = 100 - 2 \cdot 12 = 76$$



$$Q^D = 0 \rightarrow P = 50$$
$$Q^S = 0 \rightarrow P = 20/8 = 2.5$$

$$CS = \frac{76 \cdot (50 - 12)}{2} = 38 \cdot 38 = 1444$$

$$PS = \frac{76 \cdot (12 - 2.5)}{2} = 38 \cdot 9.5 = 361$$

$$TAX = 10 \longrightarrow TAX = P^D - P^S = 10$$

$$Q^D = 100 - 2P \rightarrow P^D = \frac{(100-Q)}{2}$$

$$Q^S = -20 + 8P \rightarrow P^S = \frac{Q+20}{8}$$

$$\frac{100-Q}{2} - \frac{Q+20}{8} = 10 \quad (= \text{TAX})$$

$$\rightarrow 50 - \frac{Q}{2} - \frac{Q}{8} - \frac{20}{8} = 10 \rightarrow Q = 60$$

$$P^D = \frac{100-Q}{2} = \frac{100-60}{2} = 20$$

$$P^S = \frac{Q+20}{8} = \frac{60+20}{8} = 10$$

$$PS = (10 - 2.5) \cdot \frac{60}{2} = 225$$

$$CS = (50 - 20) \cdot \frac{60}{2} = 900$$

$$DWL = 10 \cdot \frac{76-60}{2} = 80$$

$$9-2P=-2+P$$
$$11=3P$$
$$P^*=\frac{11}{3} \qquad Q^*=-2+\frac{11}{3}=\frac{5}{3}$$

$$\left.\begin{aligned} Q^D&=9-2P \\ Q^S&=-3+P \end{aligned}\right\}$$

$$9-2P=-3+P$$
$$12=3P$$
$$P^*=4$$
$$Q^*=1$$

Exercise 2

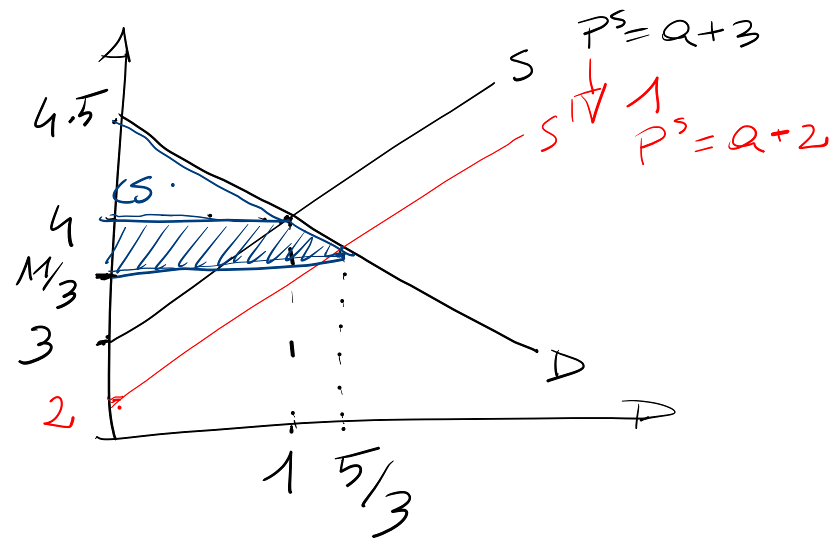
Consider a market for manufactured goods. Demand is given by $Q^D = 9 - 2P$ and supply by $Q^S = -3 + P$. The market is initially in equilibrium. The government decides to introduce a production subsidy that reduces marginal cost by 1 CHF per unit. Which of the following statements is correct? $\rightarrow Q^{S'} = -3+P+1 = -2+P$

a) Consumer surplus increases by 0.44. ✓

b) Consumer surplus decreases by 0.25 ✗

c) Producer surplus increases by 0.5.

d) Producer surplus decreases by 0.47. ✗



$$CS=\frac{4.5-4}{2}=0.25$$

$$CS'=\left(4.5-\frac{11}{3}\right)\frac{5}{3}\cdot\frac{1}{2}=\frac{25}{36}$$

$$\Delta CS = CS' - CS = \frac{25}{36}-\frac{9}{36}=\frac{16}{36}\approx 0.44$$

Exercise 3

Consider the market for organic grapes in the canton of Geneva to incentivize the production of organic wine. Quantity demanded and supplied are given respectively by $Q^D = 100 - 5P$ and $Q^S = -50 + 20P$. The state decides to grant subsidies to the farmers to target a price of CHF 3 CHF for consumers.

1. Calculate the amount spent by the government on the subsidy.

2. Calculate the revenue received by the farmers:

   a. Before the subsidy

   b. After the subsidy

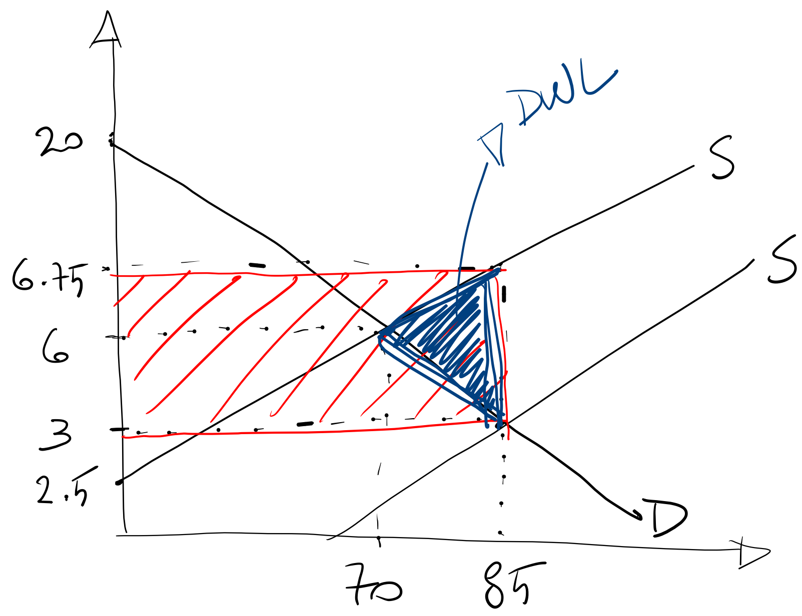
3. Calculate consumer expenditure:

   a. Before the subsidy

   b. After the subsidy

$$Q^D = 100 - 5P$$
$$Q^S = -50 + 20P$$

$$100 - 5P = -50 + 20P$$
$$150 = 25P$$
$$P^* = 6 \quad , \quad Q^* = 70$$

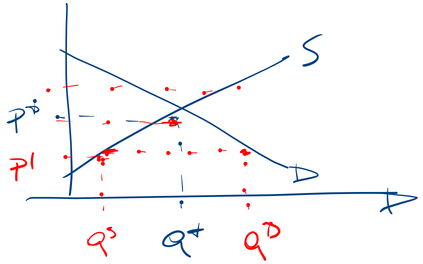


$$Q' = 100 - 5 \cdot 3 = 85$$
$$85 = -50 + 20P$$
$$135 = 20P$$
$$P^S = 6.75$$

$$SUBS = P^S - P^D = 6.75 - 3 = 3.75$$
$$TSUBS = 3.75 \cdot 85 = 318.75$$

$$R^0 = P^* \cdot Q^* = 6 \cdot 70 = 420 = C^0$$
$$R^S = P^S \cdot Q' = 6.75 \cdot 85 = 573.75 \qquad C^S = 3 \cdot 85 = 255$$

Exercise 4



shortage / excess demand

Consider the New York rental market. Demand and supply functions are given by $Q^D = 2000 - P$ and $Q^S = 0.25P$, respectively. The market is initially in competitive equilibrium. Then the government imposes a price ceiling of 1000 USD. Which of the following statements is true?

a. A shortage of 750 units is observed. ✓

b. The price ceiling increases quantity exchanged by 700 units. ✗

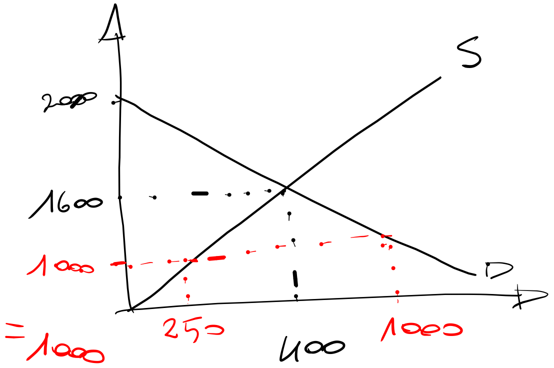
c. The price ceiling decreases quantity exchanged by 50 units. ✗

d. The price ceiling has no effect. ✗

$2{,}000 - P = 0.25P$

$\rightarrow P^* = 2000/1.25 = 1600$

$Q^* = 2000 - 1600 = 400$

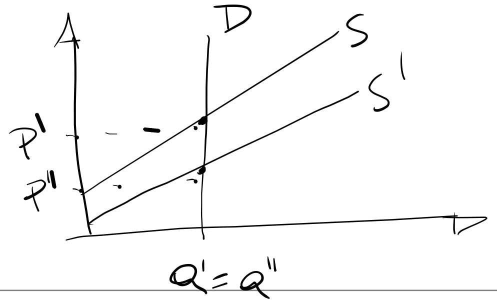


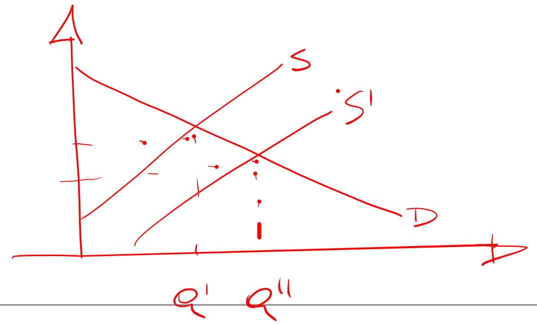
$Q^D(P=1000) = 2000 - 1000 = 1000$

$Q^S(P=1000) = 0.25 \cdot 1000 = 250$

$1000 - 250 = 750$

Exercise 5





The Swiss government decides to grant subsidies to Swiss watch producers. In equilibrium, what are the consequences of this intervention?

$SUBS = P^S - P^D$

a. The total expenditure on Swiss watches will necessarily increase. ✗

b. The consumer price increases, the producer price falls. ✗

c. The price of watches will necessarily decrease by the amount of the subsidy. ✗

d. The quantity exchanged of Swiss watches will necessarily increase if neither supply nor demand are completely inelastic. ✓

e. No change is observed in this market. ✗

$$75 - 2P = 3P$$
$$75 = 5P$$
$$P^* = 15 \qquad Q^* = 3 \cdot 15 = 45$$

Exercise 6

Consider the labor market on which demand and supply of labor is given by $Q^D = 75 - 2P$ and $Q^S = 3P$, where P is the price of labor per hour. The market is initially in equilibrium. Suppose the local government sets a minimum wage of 20 CHF per hour:

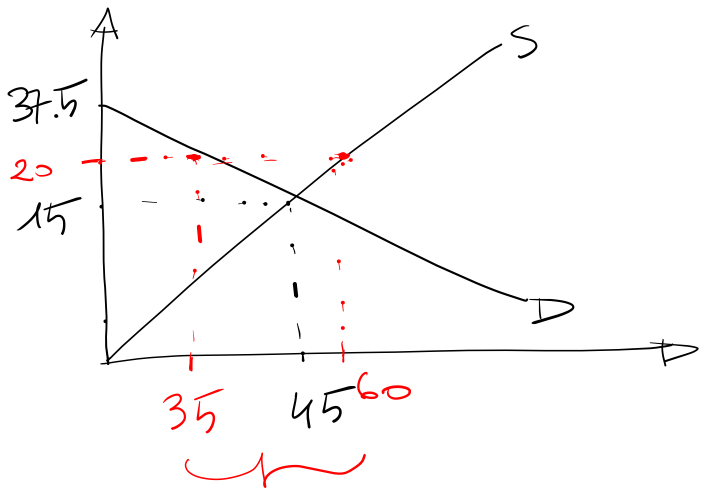
$$Q^D(P=20) = 75 - 40 = 35$$

a. An excess supply of 20 is observed. ✗

b. The quantity exchanged increases by 15. ✗

c. The quantity exchanged decreases by 10. ✓

d. No change is observed in this market. ✗



Excess supply of 25

$$Q^S(P=20) = 3 \cdot 20 = 60$$

Exercise 7

$C = (P_1 - P^*) \cdot Q_1$

$D = (P_1 - P^*) \cdot (Q^* - Q_1)\frac{1}{2}$

$C + D = (P_1 - P^*) \cdot Q_1 + (P_1 - P^*)(Q^* - Q_1)\frac{1}{2}$

$= \frac{1}{2}(P_1 - P^*)\left[2Q_1 + (Q^* - Q_1)\right]$

$= \frac{1}{2}(P_1 - P^*)(Q_1 + Q^*)$

The figure below shows the demand and supply curves for salt, and the shift in the supply curve due to the implementation of a 30% tax on the price of salt.

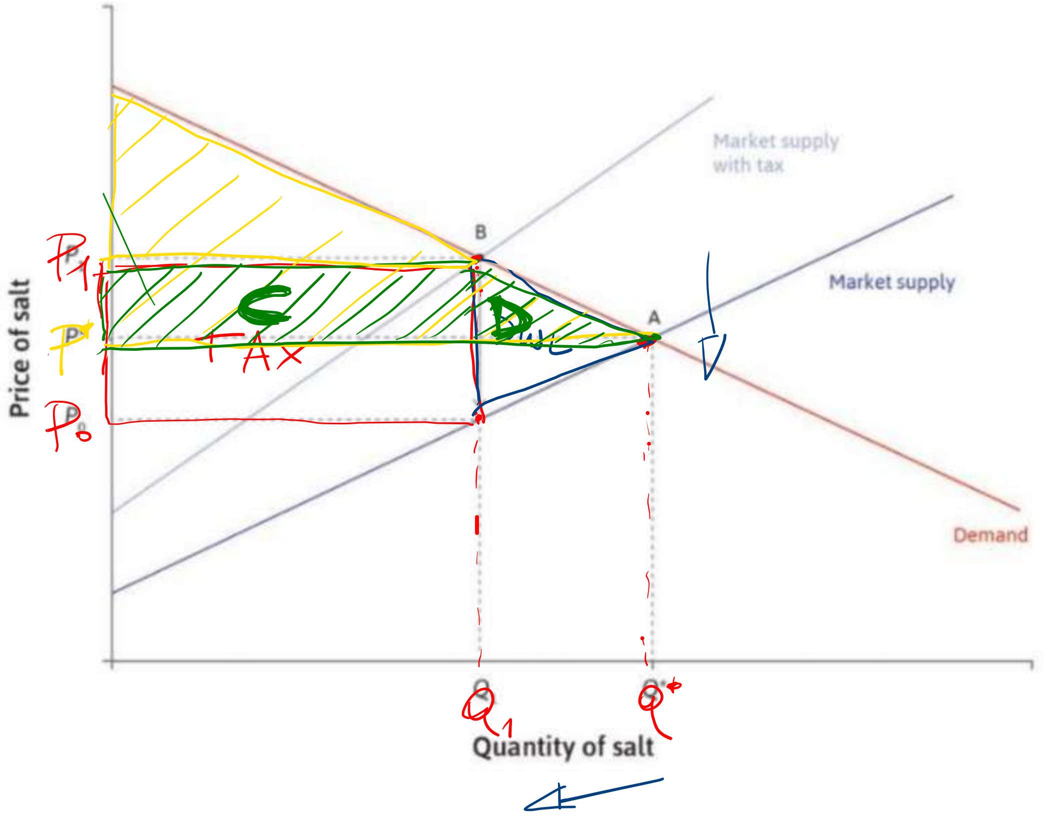
Which of the following statements are correct?

a) In the post-tax equilibrium, the consumers pay P_1 and the producers receive P^*. ✗

b) The government's tax revenue is given by $(P^* - P_0)Q_1$. ✗

c) The deadweight loss is given by $\frac{1}{2}(P_1 - P_0)(Q^* - Q_1)$. ✓

d) As a result of the tax, the consumer surplus is reduced by $\frac{1}{2}(Q^* + Q_1)(P_1 - P^*)$. ✓



Costs: basic ideas

$TC = FC + VC$

- 'Producing more costs more'
  - Agree? Disagree? → Need to hire production factors.
- 'The unit cost of production (i.e. cost per unit) rises as the volume of production rises'
  - Agree? Disagree?
  - Think about set up costs (building a plant, creating a brand or a logo, etc.)
  - Think about congestions costs (managerial costs, monitoring production units, coordination…)
  - Two kinds of "unit cost": average cost and marginal cost

$$AC = \frac{TC}{Q} = \frac{FC}{Q} + \underbrace{\frac{VC}{Q}}_{AVC}$$

Average cost and marginal cost

- Laymen think about averages.
- Economists think at the margin.

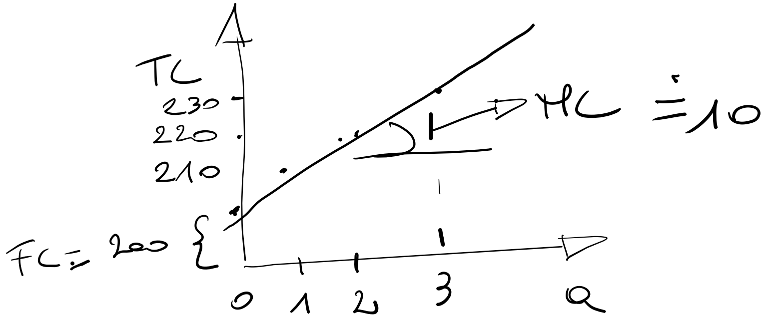
$$MC = \frac{\Delta TC}{\Delta Q} = \frac{\Delta VC}{\Delta Q}$$

$$\text{Average cost} = \frac{\text{Total cost}}{\text{Output}}$$ = the cost of <u>one</u> unit of output

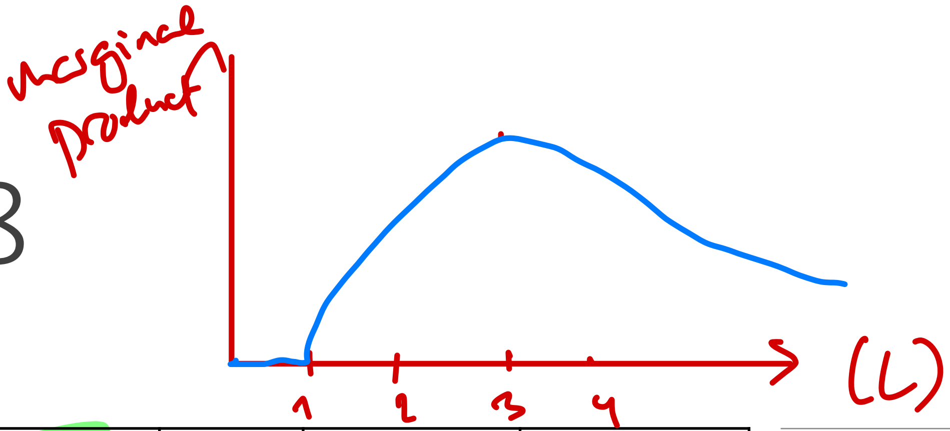
(= tangent of the angle of a point on the total cost function)

$$\text{Marginal cost} = \frac{\Delta\,\text{Total cost}}{\Delta\,\text{Output}}$$ = the cost of the <u>last</u> unit produced

(= slope of the total cost function)



Exercise 8



Cost of L : 100 CHF/day

Fixed : 200 CHF

Workers (L)	Production (q)	Marginal product (MP)	Total cost (TC)	Average cost (AC) TC/q	Marginal cost (MC)
0	0	0	200	–	–
1	20	20	300	300/20=15	(300-200)/20=5
2	50	30	400	400/50=8	(400-300)/30=3.33
3	90	40	500	5.5	100/40 =2.5
4	120	30	600	5	100/30 = 3.33
5	140	20	700	5	100/20 = 5
6	150	10	800	5.33	100/10 = 10
7	155	5	900	5.81	100/5 = 20

Clean Sweep is a company that produces brooms and sells them door-to-door. The following table describes the relationship between the number of workers hired and the production of Clean Sweep in a day.

1. Fill in the marginal product column. What shape do you see appearing? How do you explain it?

2. A worker costs 100 CHF per day and the company has fixed costs of 200 CHF. Use this information to fill in the total cost column.

3. Fill in the average cost and marginal cost columns.

4. Compare marginal product and marginal cost. Explain the relationship between the two.